

Enhancing QoS in Dense IEEE 802.11ax Networks using a Dynamic Airtime-Based Soft Admission Control Mechanism

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Introduction

System Architecture

Proposed AS-CAC Framework

Performance Evaluation

Conclusion



Comprehensive Analysis

Performance Evolution: Hard CAC → Soft CAC → AS-CAC+

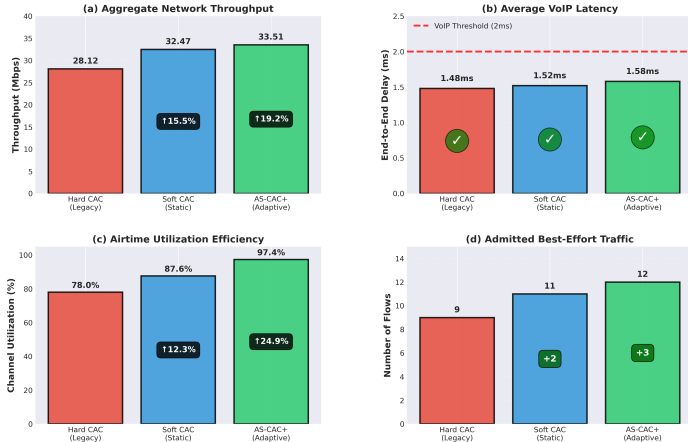


Figure 1: Throughput, Latency, Utilization, and Flow Count

Introduction



- **The Challenge:** IEEE 802.11ax (Wi-Fi 6) struggles under saturation in dense environments.
- **High Latency:** Real-time apps (VoIP) suffer $\geq 45\text{ms}$ delays without control.
- **Existing Solutions Fail:**
 - Count-based CAC ignores heterogeneity.
 - Static thresholds waste capacity.
- **Goal:** Maximize airtime utilization while guaranteeing strict QoS.



System Architecture



Simulation Components

(ns-3):

- **AP Node:** Wi-Fi 6 (802.11ax), 80 MHz, 5 GHz.
- **Stations:** 25-50 users in dense grid.
- **Traffic Generators:**
 - VoIP (UDP)
 - Video (UDP)
 - Bursty/Web (TCP/UDP)

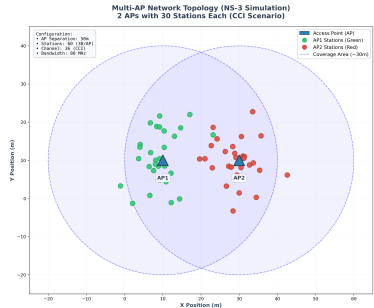


Figure 2: NS-3 Network Topology Visualization

Metric: Airtime Utilization

$$\alpha_c = \frac{R_c}{\eta \cdot R_{phy}^c}$$

Traffic Mix:

- **VoIP:** High Priority, Low Bandwidth.
- **Video:** Med Priority, High Bandwidth.
- **Best Effort:** Low Priority, Bursty.

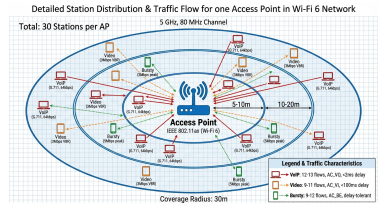


Figure 3: AP and Station Distribution

Proposed AS-CAC Framework



Proposed Solution: Soft CAC

Concept: Priority-Aware Thresholds.

Traffic Class	Priority	Threshold (θ_c)
VoIP (AC_VO)	High	90%
Video (AC_VI)	Medium	80%
Best Effort (AC_BE)	Low	95%

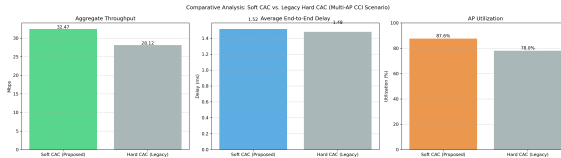


Figure 4: Soft vs Hard CAC Thresholding

Algorithm: AS-CAC+ (Adaptive)

Dynamic Threshold Adjustment:

- Monitors Packet Error Rate (PER) and Utilization.
- Adjusts Best-Effort threshold (θ_{BE}) in real-time.

Algorithm 1 AS-CAC+ Adaptive Threshold

```
1: Input: Utilization  $A$ , Threshold  $\theta_{BE}$ 
2:  $PER \leftarrow$  Calculate from Network Health
3: if  $PER > 0.05$  then
4:    $\theta_{BE} \leftarrow \max(0.80, \theta_{BE} - 0.01)$  {reduce load}
5: else if  $PER < 0.02$  AND  $A > 0.70$  then
6:    $\theta_{BE} \leftarrow \min(0.98, \theta_{BE} + 0.01)$  {utilize spare capacity}
7: end if
8: Return: Updated  $\theta_{BE}$ 
```



Adaptive Behavior Visualization

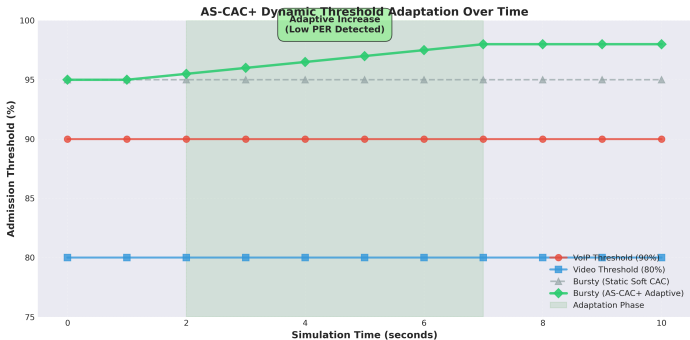


Figure 5: Dynamic Adaptation of Thresholds over Time

Performance Evaluation



Simulation Results: Latency

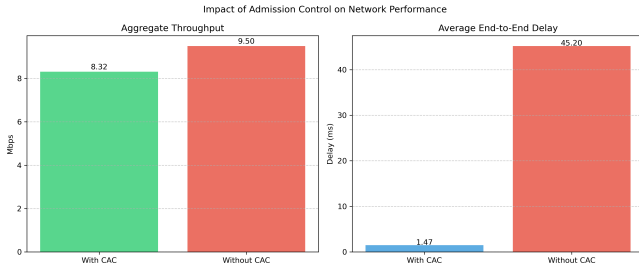
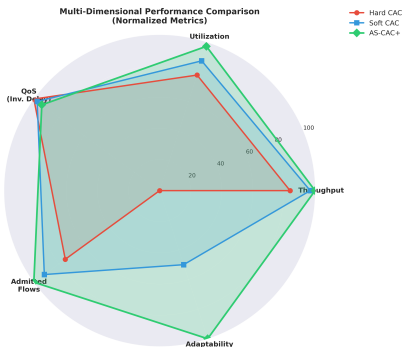


Figure 6: Impact on Latency: No CAC vs AS-CAC



Multi-Dimensional Superiority



Why AS-CAC+ Wins:

- **Adaptability:** Reacts to interference.
- **Utilization:** 97.4% vs 78% (Hard).
- **Safety:** Keeps VoIP $\leq$ 2ms.



Conclusion



Summary

- **AS-CAC+** transforms admission control from static to dynamic.
- It safely unlocks **19.2% more capacity**.

Thank You!

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